Guideline 7

Ensure moderate use of edible oils and animal foods and very less use of *ghee*/ butter/ *vanaspati*

Rationale: ***Excessive use of plant and animal based fats leads to the elevation of blood lipids thereby increasing the risk of heart disease and other illnesses.***

- ★ Fats/oils have high energy value and induce satiety.
- ★ Fats provide energy, essential fatty acids and promote absorption of fat-soluble vitamins.
- ★ Fats are precursors of biologically-active compounds in the body.
- ★ Diets that provide excess of calories, fats and cholesterol elevate blood lipids (cholesterol and triglycerides) and promote blood clotting.
- ★ Excessive fat in the diet increases the risk of obesity, heart disease, stroke and cancer.
- ★ Ill effects of excess dietary fats are initiated early in life.

Why do we need fats?

Cooking oils (liquid) and solid fats together are referred to as fats. Fats contribute to texture, flavor and taste and increase the palatability of the diet. Fats are essential for meeting some of the nutritional needs like essential fatty acids (linoleic n-6 and alpha-linolenic n-3) and serve as rich sources of energy. Therefore, fats should be consumed, in moderation. However, for the growth of young children high-calorific diets are required. This is achieved by inclusion of adequate amounts of fat (1g fat = 9 Kcals) in their diets as they cannot consume large quantities of bulky cereal - pulse- based diets.

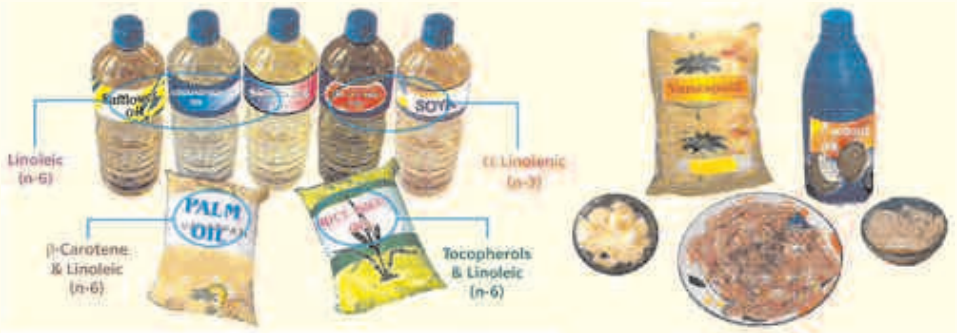
Fats also promote the absorption of the four fat-soluble vitamins (A,D,E and K), impart a feeling of fullness and satisfaction and thus, delay the onset of hunger.

Along with proteins, fats constitute major components of body fluids and cell membranes. The two essential fatty acids (EFA) namely, linoleic (LA n-6) and alpha-linolenic (ALA n-3) acids (important dietary polyunsaturated fatty acids) are metabolized at various sites in the body to generate a group of biologically-active compounds, which perform several important physiological functions.

What are the sources of fat?

Dietary fats can be derived from plant and animal sources. Fats that are used as such at the table or during cooking (vegetable oils, vanaspati, butter and ghee)

are termed as "visible" fats. Fats that are present as an integral components of various foods are referred to as "invisible" fat. Fats, in processed and ready to eat foods are known as hidden fats. Cereals contain only 2-3% of invisible fat. However, their contribution to overall fat intake is significant as they contribute to bulk of our Indian diets. The small amounts of invisible fat present in various foods add up to a substantial level in our daily diet (about 15 g in rural population and 30g among urban middle-income and high-income groups). Most animal foods provide high amounts of invisible fat.



How much visible fat do we need?

The total fat (visible + invisible) in the diet should provide between 20-30% of total calories. The visible fat intake in the diets can go upto 50g/person/day based on the level of physical activity and physiological status. Adults with sedentary lifestyle should consume about 25 g of visible fat, while individuals involved in hard physical work require 30 - 40g of visible fat. Visible fat intake should be increased during pregnancy and lactation to 30g. The higher fat and EFA requirements during pregnancy and lactation are to meet the requirements of fetus and young infants, in view of their crucial role in physical and neuronal growth and development. Diets of young children and adolescents should contain about 30-50g/day. However, ingestion of too much fat is not conducive to good health.

What are the chemical components of fat?

Fatty acids: All fats in foods provide mixtures of three types of fatty acids, which are the "building blocks" of fats. Fatty acids are the primary constituents of all dietary fats. Based on their chemical nature, the fatty acids are broadly grouped as saturated (SFA), monounsaturated (MUFA) and polyunsaturated (PUFA). There are several fatty acids in each group. Fats from coconut oil, *vanaspati*, animal fats (ghee and butter) and animal foods like milk, milk products and meat provide saturated fatty acids. The short and medium chain saturated fatty acids present in ghee, butter

and coconut oil are easily digested and absorbed and are therefore, good for infants and young children. However, high intake of saturated fatty acids increases atherogenic risk and their intake should be limited in adults. Oils from sources such as palm, groundnut, cottonseed, sesame and olive are rich in monounsaturated fattyacids as compared to other oils. Linoleic (n-6) and a-linolenic (n-3) acids are the simple PUFA, which are present only in plant foods (Table 6). All vegetable oils (except coconut) are good sources of linoleic(n-6) acid. Soyabean, rapeseed and mustard oils are the only vegetable oils, which contribute significant proportion of a-linolenic (n-3) acid. Legumes/pulses mustard and fenugreek seeds and green leafy vegetables are also good sources of a-linolenic (n-3) acid (Table 7). On the other hand, fish and fish oils provide long chain n-3 fatty acids, which are biologically more active than alpha-linolenic (n-3) acid present in plant foods.

Table - 6
Major Types of Fatty Acids in Fats and Oils

SATURATED	MONO-UNSATURATED	POLYUNSATURATED		
		LINOLEIC (n-6)		a –LINOLENIC (n-3)
Coconut Palm kernel Ghee/butter Vanaspati	Red palm oil Palmolein Groundnut Ricebran Sesame	**Low**	Red palm oil Palmolein	Rapeseed, Mustard Soyabean
		Moderate	Groundnut, Ricebran Sesame	
		High	Safflower,Sunflower Cottonseed, Corn, Soyabean	

Dietary fats also contain minor components such as tocopherols, tocotrienols, sterols etc. The natural flavour of fats/oils is largely due to these minor components. Since most of the minor components are antioxidants, they prevent fats from going rancid. Tocotrienols in palm oil, lignans in sesame oil and oryzanol and tocotrienols in rice-bran oil reduce blood cholesterol. Refining of oils, though does not alter their fatty acid composition, modifies the composition of minor components; for example, carotenes are lost during refining of crude palm oil.

Cholesterol: Cholesterol is present only in foods of animal origin such as milk, meat, shrimp and prawn, but not in plant foods. Vegetable oils do not contain cholesterol. Egg yolk, and organ meats such as liver, kidney and brain contain very high amounts of cholesterol. Cholesterol is found in all body cells and plays a key role in the formation of brain, nerve tissue and is a pre-cursor for some hormones and vitamin D. It is synthesized in the body and hence it is not an essential dietary component.

Table – 7
Quantities of foods required to furnish 0.1 g ALA

Foods	Gram
Cereal/Millet	
Wheat & Pearl millet (bajra)	70
Pulses	
Blackgram (kala chana), kidney beans (rajmah) & cowpea (lobia)	20
Other pulses	60
Vegetables	
Green leafy	60
Other Vegetables	400
Fruits	400
Spices	
Fenugreek seed (methi)	5
Mustard (sarson)	1
Unconventional	
Flaxseed (alsi)	0.5
Perilla seeds (Bhanjira)	0.3

Higher dietary cholesterol increases blood cholesterol. The blood cholesterol-elevating effect of dietary saturated fats increases, when cholesterol consumption is high. Therefore, cholesterol intake should be maintained below 200 mg/day. One can reduce both saturated fat and cholesterol intake by limiting the consumption of high-fat animal foods like butter, ghee, meat, egg and organ meats and consuming low fat (skimmed) milk instead of whole milk. However, consumption of eggs (3eggs/week) is recommended in view of several nutritional advantages.

What are the physiological/health implications of different fats/fatty acids?

Saturated fatty acids are known to increase serum total and LDL-cholesterol levels, reduce insulin sensitivity and enhance thrombogenicity and increase CVD risk. Therefore, SFA intake should not exceed 8-10% of total energy. Skimmed milk should be preferred instead of whole milk. Restrict consumption of butter and cheese. PUFAs are essential components of cell membranes. While n-6 PUFAs are predominant in all cells, the nerve tissue has high levels of long chain n-3 PUFA. An appropriate balance of the these two classes of PUFAs, namely, linoleic and alpha-linolenic acids in the diets is essential for the functioning of vascular, immune, nervous and renal systems and for early human development. PUFAs particularly n-3 PUFA increase insulin sensitivity, increase peripheral glucose utilization and decrease adiposity and hence are anti-atherogenic. n-6 PUFA decrease plasma

LDL - Bad Cholestrol (SFA)
HDL - Good Cholestrol

cholesterol as well as HDL cholesterol level (only at high intake). The lipid lowering and other physiological effects of individual members of the PUFAs vary widely. As compared to linoleic acid, alpha-linolenic (n-3) acid is more beneficial for prevention of inflammation and accumulation of fatty material in blood vessels (altheroscleros) and clotting of blood (thrombosis). The long chain n-3 PUFA of fish oils and micro algae have greater antiatherogenic, antithrombotic and anti-inflammatory effects than alpha-linolenic (n-3) acid of plant foods. They are important for vision and brain growth. Therefore, pregnant women should consume foods that are rich in ALA and long chain n-3 PUFA from fish and fish oils.

The intake of PUFA should be 8-10% of energy intake. The remaining 8-10% of fat calories can be derived from mono-unsaturated fatty acids, which also help in maintaining plasma cholesterol. Excessive use of highly unsaturated fats should be avoided. Further, to get a good proportion of all the classes of fatty acids, it is advisable to consume more than one type of vegetable oils.

Fats/ lipids (triglycerides, cholesterol and phospholipids) are transported in blood in combination with proteins in the form of 'lipoproteins'. The low density lipoproteins (LDL) transport cholesterol from liver to various tissues. High blood levels of LDL cholesterol ('bad' cholesterol) result in accumulation of lipids in the cells (atherogenic effect). High density lipoproteins (HDL) ('good' cholesterol) transport excess cholesterol from the tissues to the liver for degradation, and are therefore, anti-atherogenic.

Choice of cooking oils

In view of the above, an ideal quality fat for good health is the one which maintains a balance, so as to give a ratio of polyunsaturated/ saturated (PUFA/ SFA) of 0.8-1.0, and linoleic/ a-linolenic (n-6/ n-3) of 5-10 in the total diet. For ensuring this appropriate balance of fatty acids in cereal-based diets, it is necessary to increase the a-linolenic (n-3) acid intake and reduce the quantity of linoleic (n-6) acid obtained from the cooking oil. Hence, the choice of cooking oil should be as follows:

Groundnut or Sesame or Rice bran + Mustard
Groundnut or Sesame or Rice bran + Canola
Groundnut or Sesame or Rice bran + Soyabean
Palmolein + Soyabean
Safflower or Sunflower + Palmolein + Mustard

Use of more than one source of fat/oil has the added advantage of providing a variety of minor components in the diet. An additional way of increasing alpha-linolenic (n-3) acid intake is to ensure regular consumption of oils and foods rich in

alpha-linolenic (n-3) acid (Table 6 & 7). Non-vegetarians have an advantage of eating fish, which provides preformed long chain n-3 PUFA. Ideally, part of visible fat and/or invisible fat from animal foods may be substituted by whole nuts and legumes with good proportion of -linolenic (n-3) acid, which are also good sources of protein, fiber, vitamins and minerals (ALA content of foods is given in annexure 9).

The plant oils in addition contain certain useful substances such as lignans (sesame oil), sterols, tocopherols (vitamin E) oryzanole (rice bran oil), carotenoids - all of which reduce cholesterol and reduce oxidant damage due to ageing, inflamation which occur in chronic diseases.

What about *vanaspati*?

Vanaspati is prepared by hydrogenation of vegetable oils. During hydrogenation, the liquid oils become solid because the mono and polyunsaturated fatty acids are converted into saturated fatty acids and isomers called *trans* fatty acids. *Vanaspati* is used as a substitute for ghee in cooking and the preparation of bakery products, sweets and snack foods. Since, saturated fats are resistant to oxidation, foods prepared in *vanaspati* keep fresh for a longer period. Current evidence indicates that saturated fatty acids and a high intake of *trans* fatty acids may increase the risk of heart disease. Therefore, it is essential to limit the intake of *vanaspati*. The intake of *trans* fatty acids should not exceed 1% of energy intake.

POINTS TO PONDER

- **Take just enough fat.**
- **Substitute part of visible fat and invisible fat from animal foods with whole nuts.**
- **Moderate the use of animal foods containing high fat, SFA and cholesterol.**
- **Limit the use of ghee, butter, especially *vanaspati* as a cooking oil.**
- **Choose low- fat dairy foods in place of regular whole fat dairy foods.**
- **Eat foods rich in alpha-linolenic (ALA) acid such as legumes, green leafy vegetables, fenugreek and mustard seeds.**
- **Eat fish more frequently (at least 100-200g /week), prefer it to meat, poultry and limit/ avoid organ meats such as liver, kidney, brain etc.**
- **Egg has several important nutrients but is high in cholesterol. Limit the consumption to 3 eggs/ week. However, egg white may be consumed in good amounts.**
- **Minimize consumption of ready- to- eat fast foods, bakery foods and processed foods prepared in hydrogenated fat.**
- **Use of re-heated fats and oils should be avoided.**
- **Use fats and oils in moderation and consume varieties of foods to get good proportion of all fatty acids for optimal health benefits.**